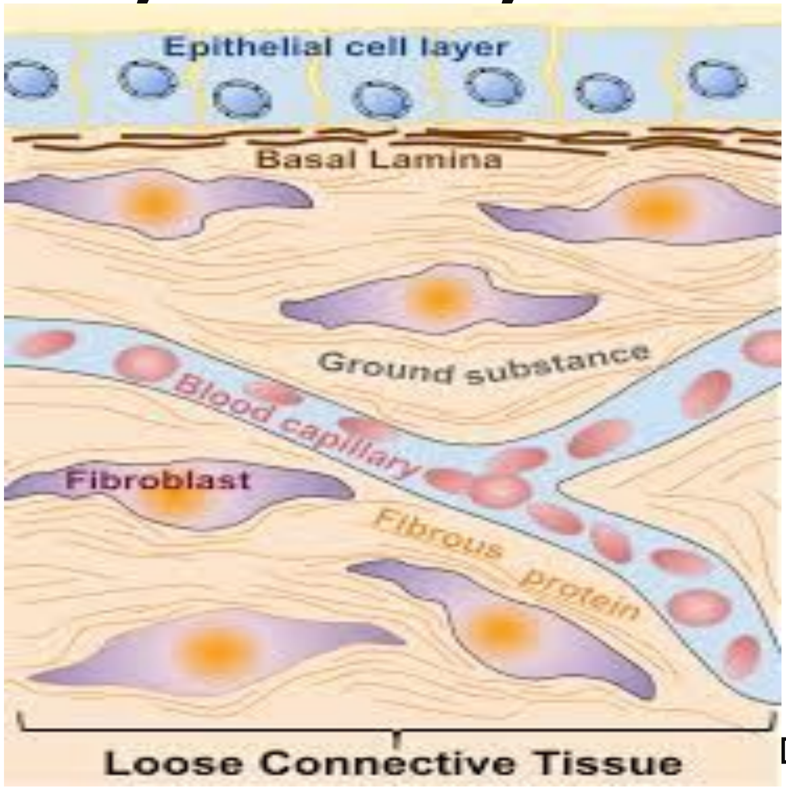
THE EXTRACELLULAR MATRIX
&
SPECIALIZED CONNECTIVE TISSUES OF BONES, GUMS, AND TEETH.



Dr. Humaira Aman Ali

LEARNING OBJECTIVES

- Understand the structure and functions of the extracellular matrix (ECM).
- Explain major components of ECM.
- Describe heteropolysaccharides and glycosaminoglycans (GAGs).
- Relate ECM biochemistry to dental tissues and clinical relevance.

CONNECTIVE TISSUE

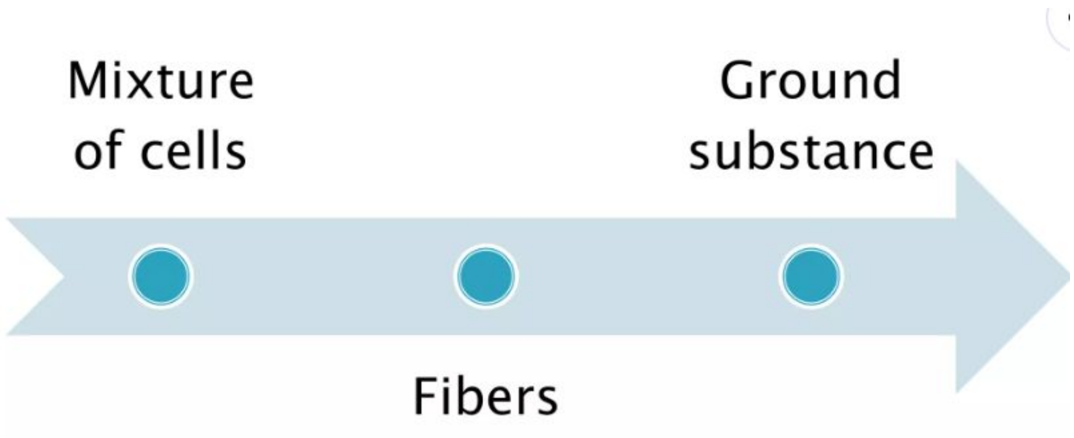
The connective tissue is formed by:

CELLS

and

EXTRACELLULAR MATRIX

(intercellular matrix)



CELLS OF CONNECTIVE TISSUE

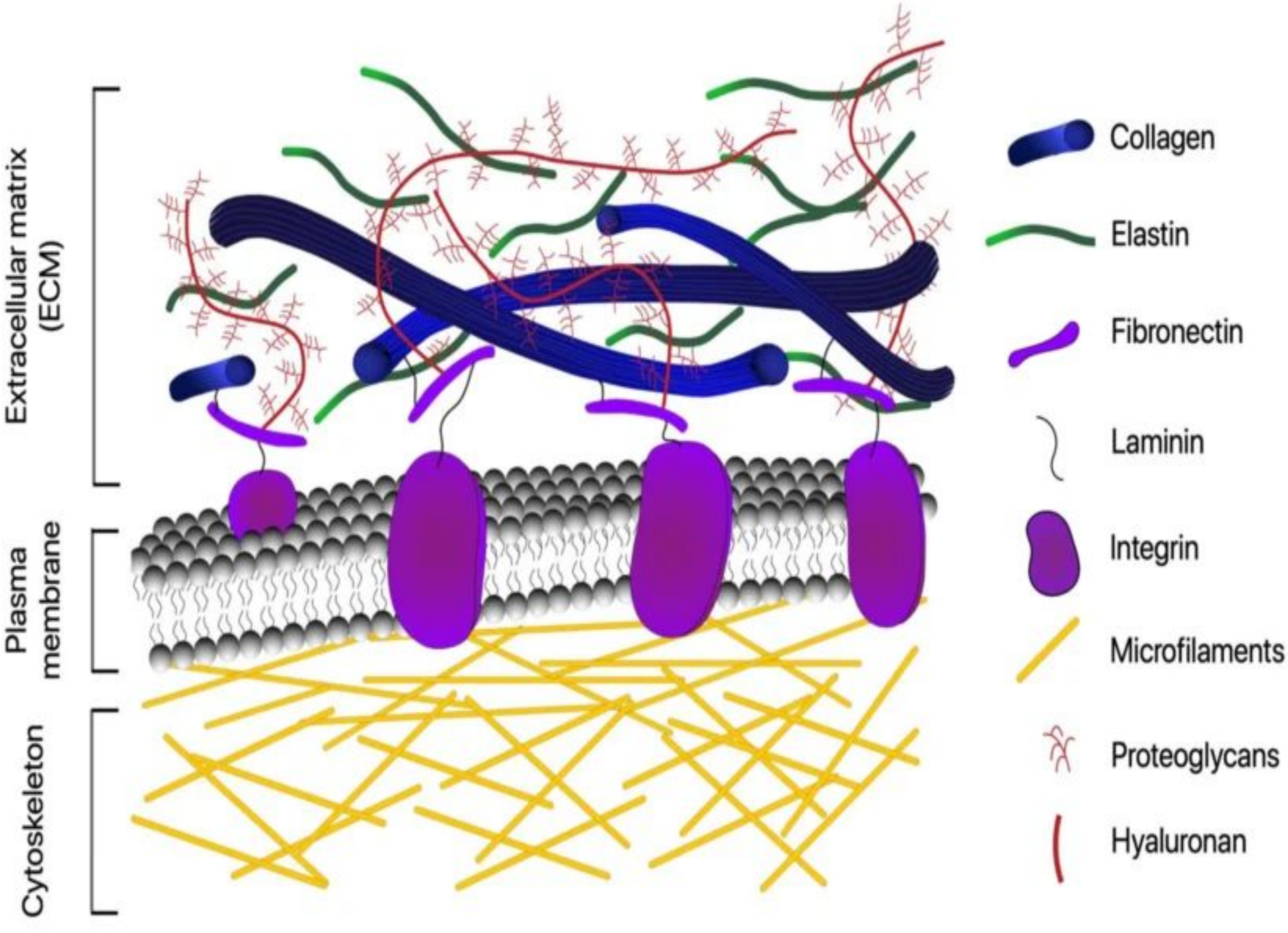
- Fibroblasts
- Chondroblasts (cartilage)
- Osteoblasts (bone)
- Odontoblasts (tooth)

These cells synthetise extracellular matrix.

WHAT IS THE EXTRACELLULAR MATRIX?

Definition:

- “The extracellular matrix is **a complex network of proteins, glycoproteins, and polysaccharides** that provide structural and biochemical support to surrounding tissues”.


Extracellular matrix
(ECM)
Plasma
membrane
Cytoskeleton
Collagen
Elastin
Fibronectin
Laminin
Integrin
Microfilaments
Proteoglycans
Hyaluronan

EXTRACELLULAR MATRIX

- Complex network of macromolecules outside cells.
- Provides structural support and biochemical signaling.
- Essential for tissue integrity and cell communication.
- Highly important in connective tissues and dental structures.

PARTS OF THE EXTRACELLULAR MATRIX

- FIBRILLAR PROTEINS (collagen, elastin)
    - insoluble in water
- GLYCOPROTEINS (e.g. fibronectin,laminin)
- GLYCOSAMINOGLYCANS AND PROTEOGLYCANS (ground substance)
    - soluble in water, easily hydratated

Saccharide content increases

FUNCTIONS OF MAJOR COMPONENTS OF ECM

Components of Interstitial matrix- ECM

FIBROUS STRUCTURAL PROTEINS Confers tensile strength and recoil.	• Collagen • Elastin
WATER HYDRATED GELS Permits compressive resistance and lubrication	• Proteoglycans • Hyaluronans
ADHESIVE GLYCOPROTEINS Connects ECM elements to one another and to cells.	• Fibronectin • Laminin

FIBRILLARY PROTEINS

- Structural proteins
  - Collage
    n
  - Elastin

firmness
elasticity

COLLAGEN

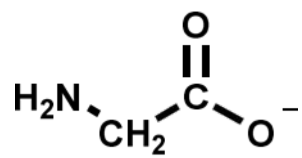
- Most abundant protein in the human body. They form approximately 25 % of all body proteins.
- Main protein of the extracellular matrix.
- Component of tendons, cartilages, bones, and teeth (dentin and cement), skin and vessels.
- Provides tensile strength.
- From the nutritional point of view not fully valuable or an incomplete protein, b/c it doesn't contain tryptophan.

PRIMARY STRUCTURE OF COLLAGEN

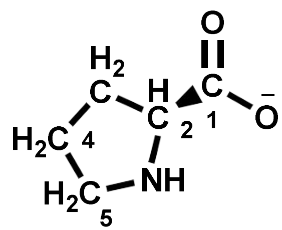
- Triple helix structure rich in glycine, proline, hydroxyproline.

Fundamental AA

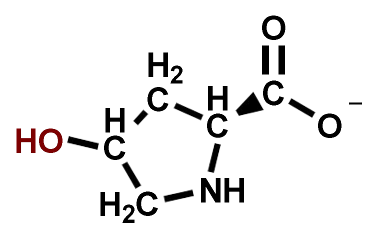
- Glycine



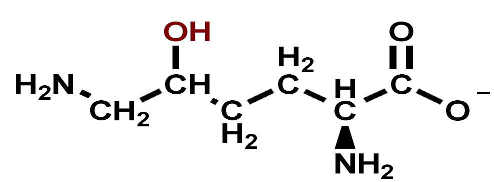
- Proline

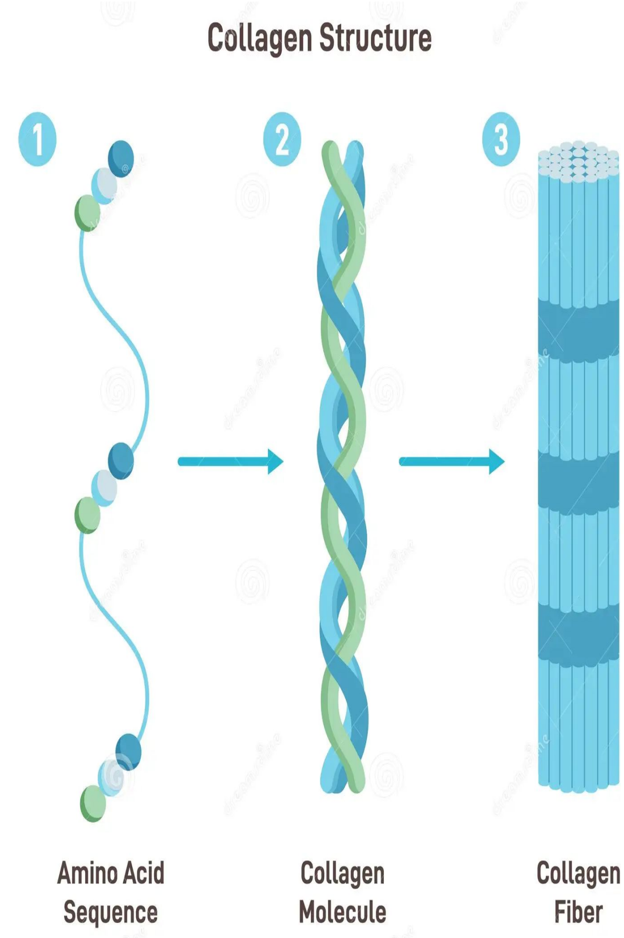


- Derived AA
  - 4-Hydroxyproline



  - 5-Hydroxylysine





CONT..

Characteristic AA sequence

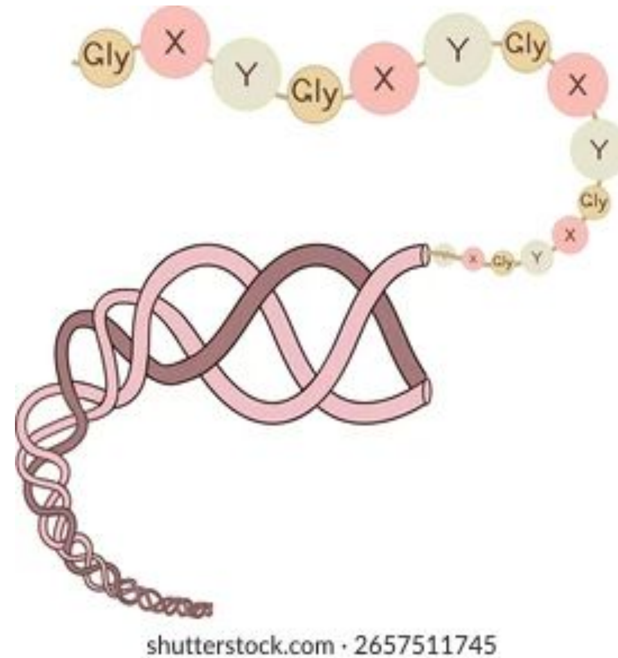
Triplet

- Gly - X - Y -

Glycine

Proline

Hydroxyproline
(Hydroxylysine)



- A repetitive Gly-X-Y pattern, where Glycine (Gly) appears at every third position.
- Proline (Pro) often occupies the X position, and Hydroxyproline (Hyp) frequently occupies the Y position.

This structural motif (Gly-Pro-X or Gly-X-Hyp) enables the formation of a tight triple helix.

SECONDARY STRUCTURE OF COLLAGEN

Comparison of collagen helix to the α-helix.

Collagen helix

- levorotatory helix
- 3 AA/turn
- intrachain hydrogen bonds not present

α -helix (the most common secondary structure in proteins)

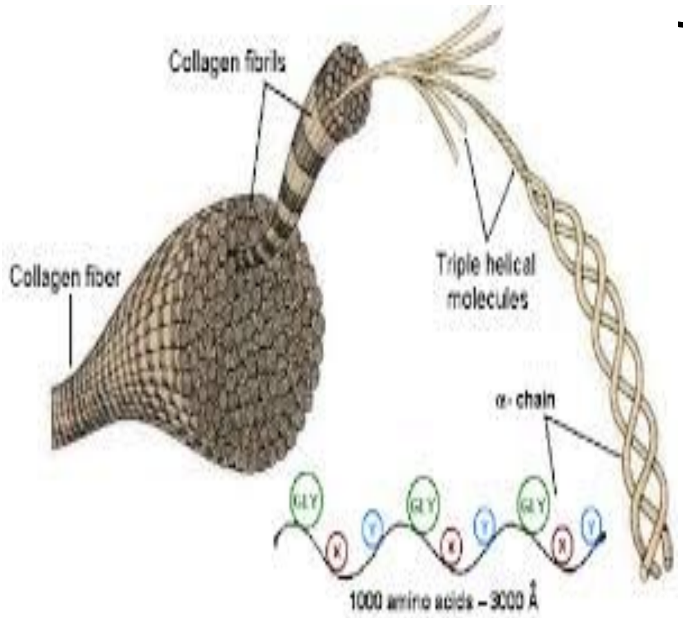
- dextrorotatory helix
- 3.6 AA/turn
- stabilization by intrachain hydrogen bonds

Alpha helix

Alpha-helix	Collagen triple helix
Single chain	Three chains
Globular proteins & fibrous proteins (e.g. keratin)	Collagen
H-bonds *within* chain	H-bonds *between* chains
3.6 residues/turn	3 residues/turn
Right-handed helix	Left-handed helix
	Right-handed superhelix
Almost any sequence (NOT proline)	- Gly - X - Y - Gly - X - Y - X= mainly proline Y= mainly hydroxy-proline

Collagen helix

TRIPLE HELIX

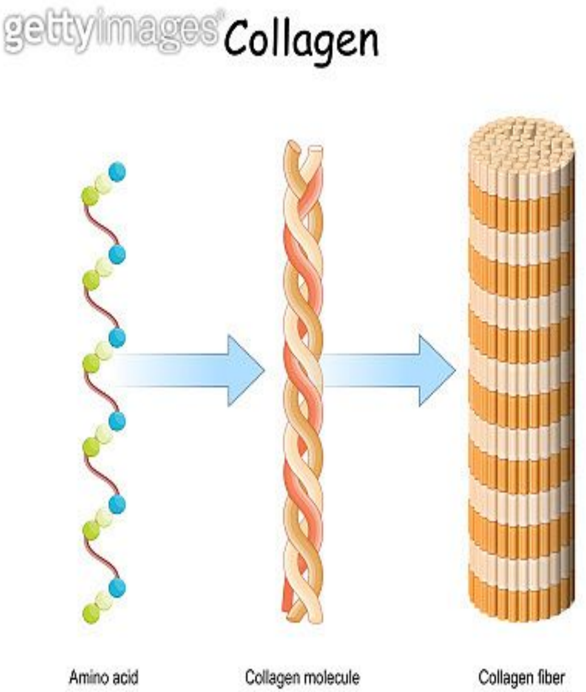


Three α-chains of collagen

↓

Triple helix

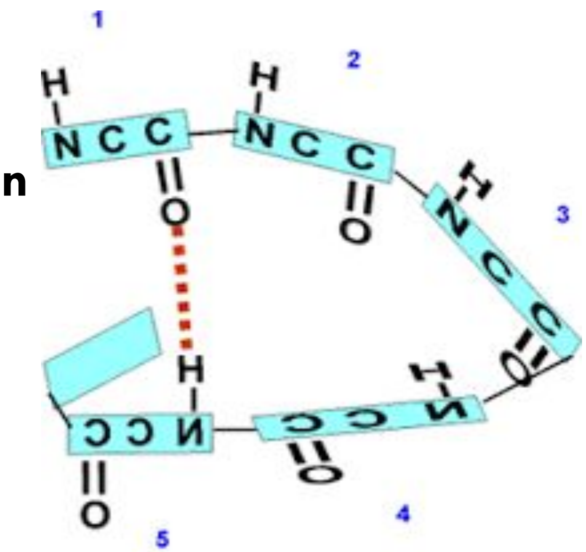
Relatively rigid



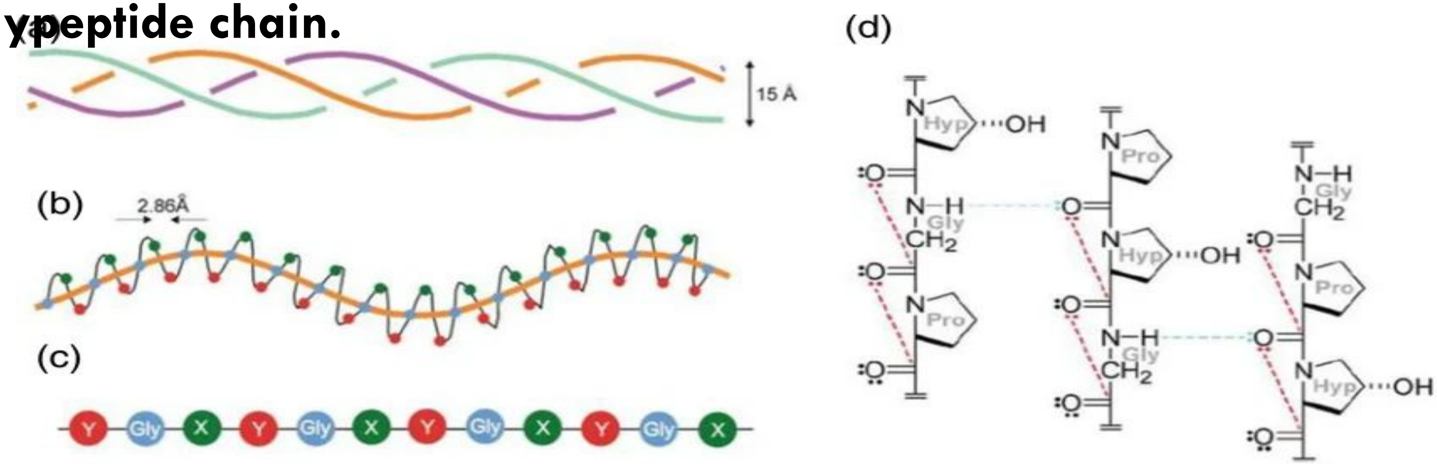
This structure is responsible for the tensile strength.

HYDROGEN BONDS IN ALPHA HELIX VS TRIPLE HELIX

Intramolecular hydrogen bonds that form between the carbonyl oxygen of one amino acid and the amide hydrogen of another, typically four residues ahead (i + 4 pattern).



Triple helix is stabilized by hydrogen bonds between peptide bond -NH group of glycine and C=O group of the peptide bond of the adjacent polypeptide chain.



COLLAGEN CHAINS

The collagen chain is extraordinarly long.

The collagen chains are called $\alpha 1 - \alpha 3$.

- They differ in AA representation
- Products of different genes – e.g. $\alpha_{1(I)}$ or $\alpha_{2(V)}$
- Roman digit labels the collagen type

More than 30 different types of collagen exists.

COLLAGEN CHAINS

The representation of chains differs in individual types of collagens.

The collagens may form homotrimers or heterotrimers.

Homotrimers

- molecule of collagen is formed by three identical chains;
- e.g. collagen type III is formed by three $\alpha_{1(III)}$ chains

Heterotrimers

- molecule of collagen is formed by different chains;
- e.g. collagen type I is assembled of two $\alpha_1(I)$ $_{\text{chains}}$ and one $\alpha_2(I)$ $_{\text{chain}}$

COLLAGEN SYNTHESIS

Collagen is an example of a protein, whose synthesis is connected with many posttranslational modifications (treatment of the polypeptide chain), which take part intra- and extracellularly.

SYNTHESIS AND POSTTRANSLATIONAL MODIFICATIONS OF COLLAGEN

Intracellular processes

Synthesis of polypeptide chain
Hydroxylation of proline and some lysine residues

Glycosylation of selected hydroxylysine residues

Formation of –S-S- bonds in extension peptides

Triple helix formation

Secretion of procollagen

POSTTRANSLATIONAL MODIFICATIONS OF COLLAGEN

Extracellular processes

Proteolytic removal of propeptides

Assembling of collagen fibrils

Formation of cross-links

Posttranslational modifications in the course of collagen synthesis

INTRACELLULAR
PROCESSES

HYDROXYLATION OF PROLINE AND LYSINE RESIDUES

Enzymatically catalyzed reaction

- Prolylhydroxylase
- Lysylhydroxylase

Dioxygenases contain Fe

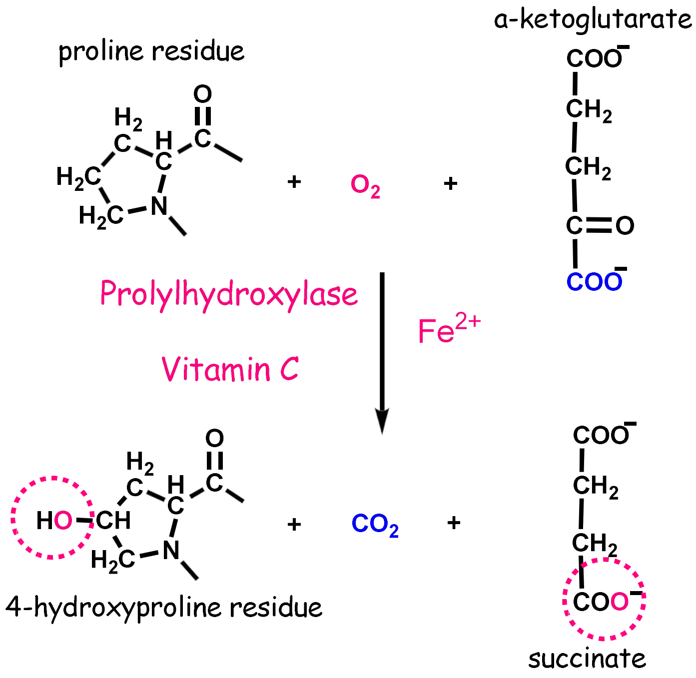
cofactors

- Vitamin C !!!
- α-ketoglutarate

Reaction needs oxygen. One O atom forms -OH group of hydroxyproline, the other becomes part of the originating succinate.

HYDROXYLATION OF THE PROLINE AND LYSINE RESIDUES

Reactions catalyzed by prolylhydroxylase



Dioxygenase contains Fe

Vitamin C Maintains Fe^{2+} in a reduced state

HYDROXYLATION OF PROLINE AND LYSINE

Reaction catalyzed by prolylhydroxylase

- reaction highly specific only for proline attached in the polypeptide chain to the amino group of the glycine

N-end -Gly-Pro-Ser-Gly-Pro-Pro-Gly-Leu- C-end

× ×

Hydroxylation

-Gly-Pro-Ser-Gly-Pro-Hyp-Gly-Leu-

HYDROXYLATION OF PROLINE AND LYSINE

Importance of proline and lysine residues hydroxylation

Hydroxyproline

- necessary for origin of triple helix by formation of hydrogen bonds between individual chains

Hydroxylysine

- glycosylation on the formed -OH group

DEFICIENCY OF VITAMIN C

Nonhydroxylated chain is not able to mature

The stable triple helix cannot be formed

Immediate degradation inside the cell

Loss of collagen in the matrix

Falling out of teeth
Vascular fragility
Poor wound healing

VITAMIN C DEFICIENCY

Avitaminosis - scurvy

Manifestation of avitaminosis in oral cavity

- swollen reddish gums
- falling out of the teeth

GLYCOSYLATION

- Attachment of galactose or galactosylglucose to -OH group of the hydroxylysine

Enzymatically catalyzed reaction

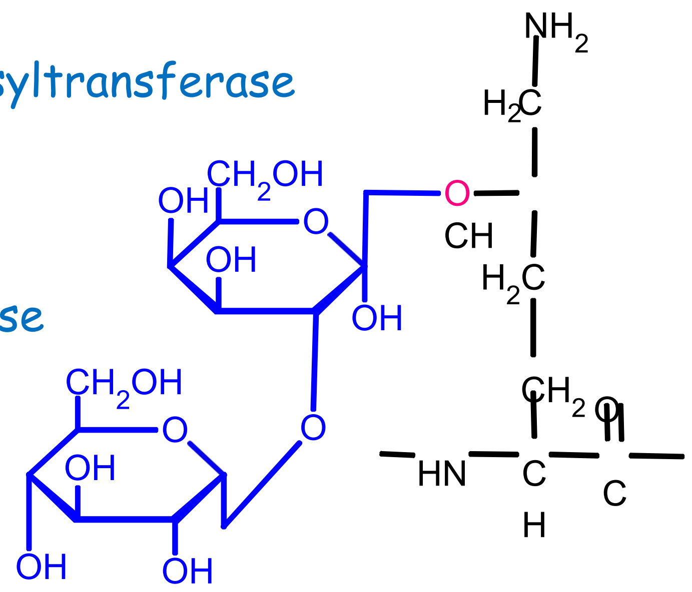
- Galactosyltransferase
- Glucosyltransferase

GLYCOSYLATION

Glycosylated residue of hydroxylysine in the molecule of collagen

Galactosyltransferase

Glucosyltransferase



β-glycoside bond

FORMATION OF –S-S- BONDS

Disulphide bonds

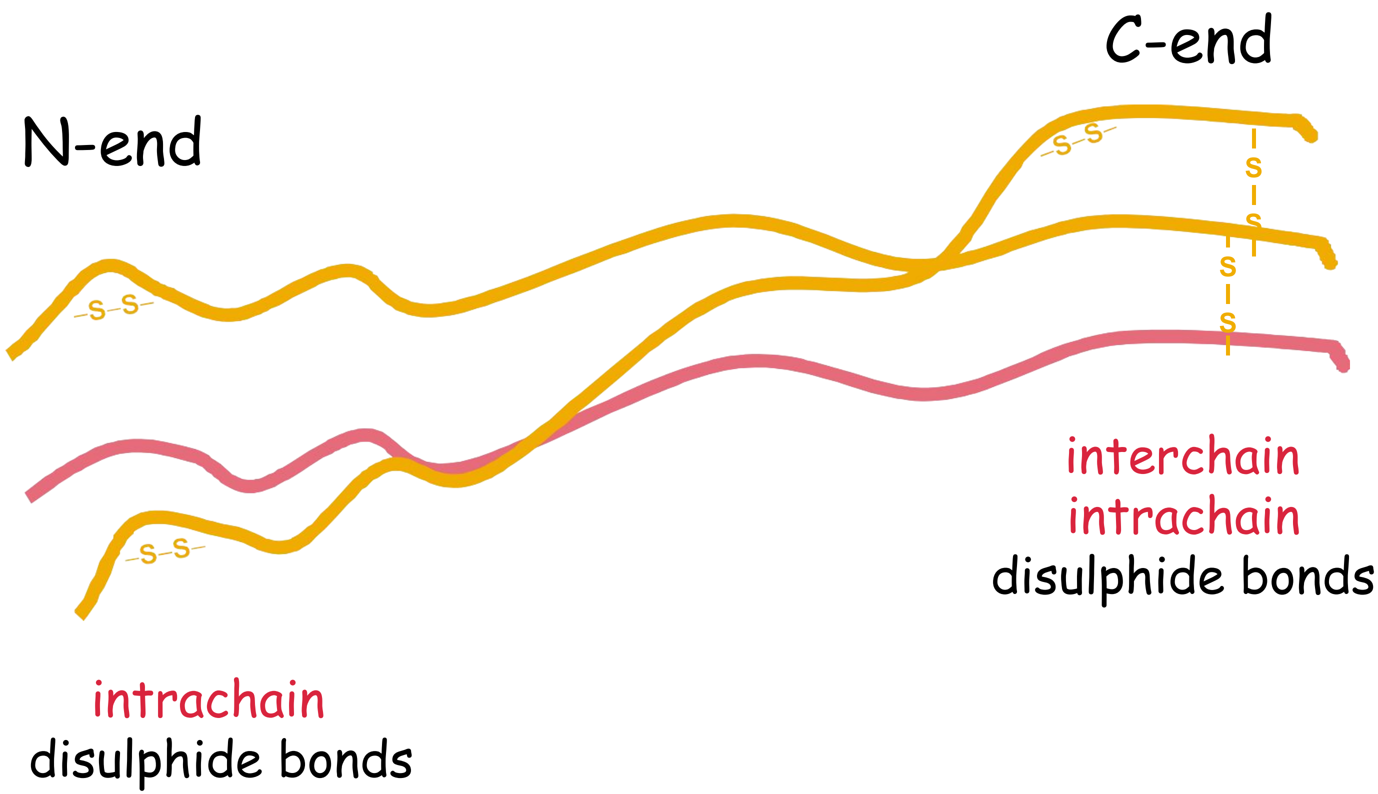
- in the region of C-terminal propeptides
  - interchain and intrachain disulphide bonds

- in the region of N-terminal propeptides
  - intrachain disulphide bonds

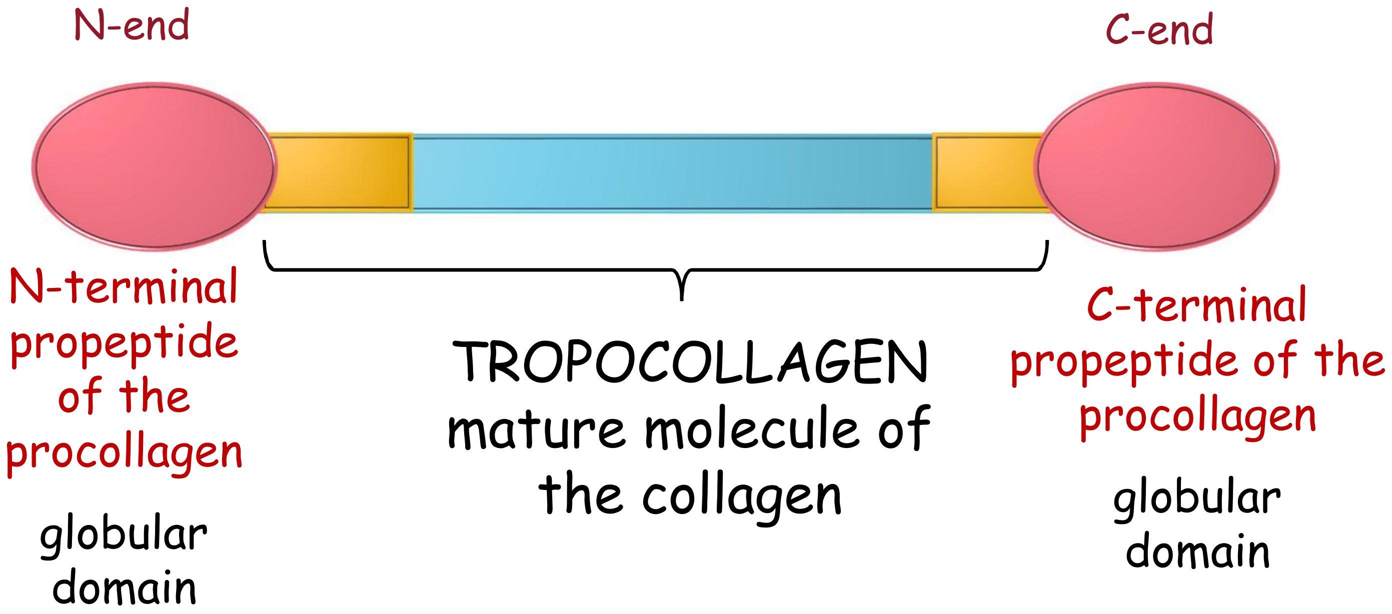
Importance

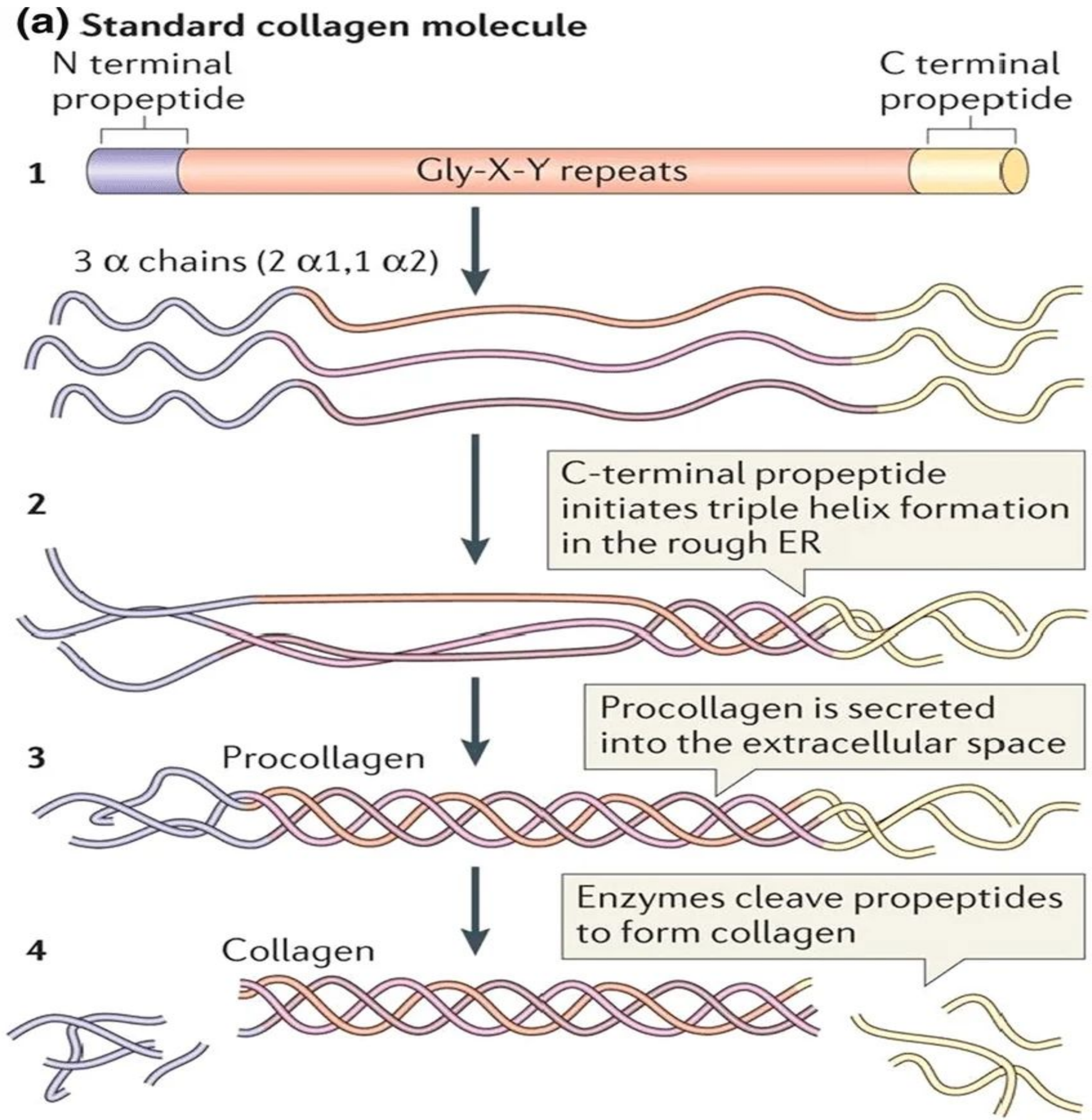
- necessary for initiation of triple helix formation
  starts from the C-end
- secretion out of the cell

FORMATION OF –S-S- BONDS



PROCOLLAGEN

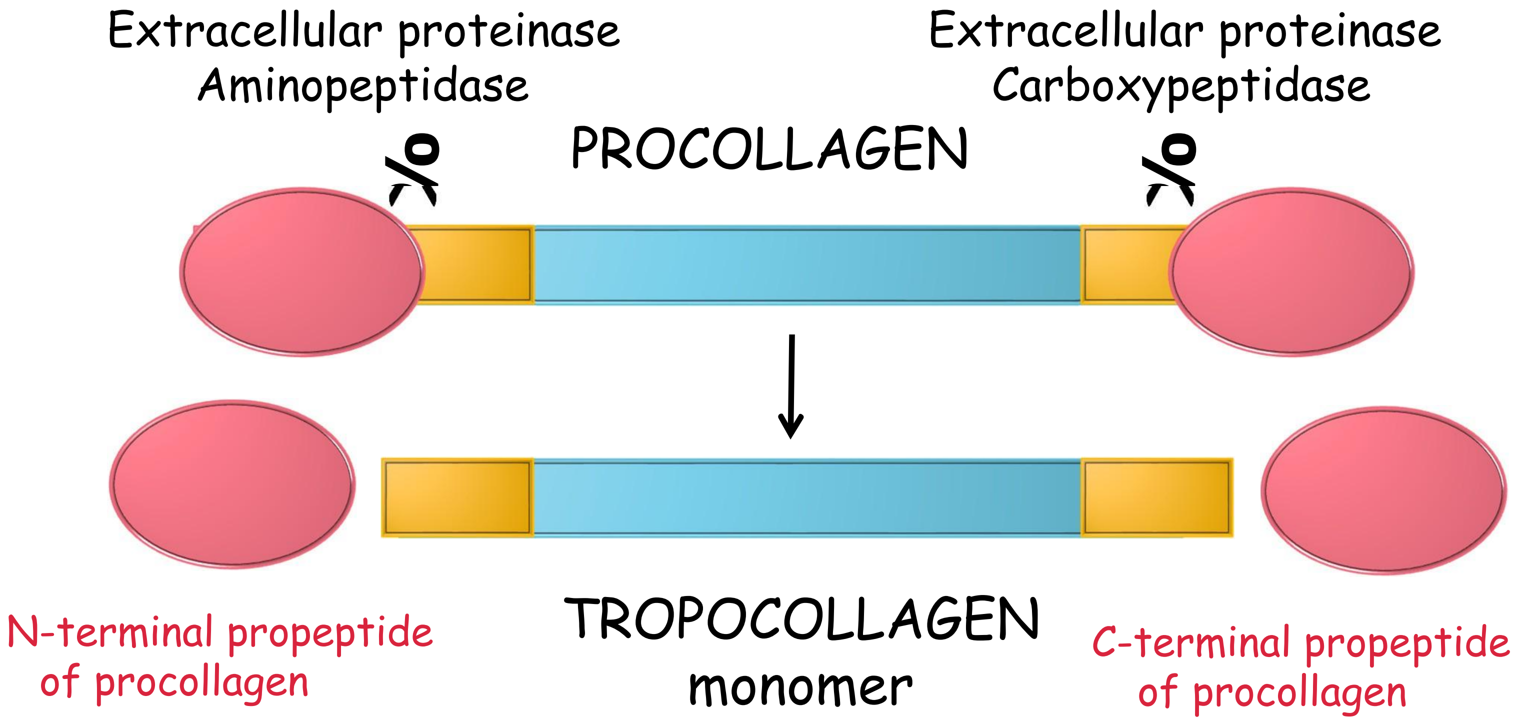



(a) Standard collagen molecule
N terminal propeptide
C terminal propeptide
1
Gly-X-Y repeats
3 α chains (2 α1,1 α2)
2
C-terminal propeptide initiates triple helix formation in the rough ER
3
Procollagen
Procollagen is secreted into the extracellular space
4
Collagen
Enzymes cleave propeptides to form collagen

Posttranslational modifications in the process of collagen synthesis

EXTRACELLULAR PROCESSES

CLEAVING OF THE PROPEPTIDES

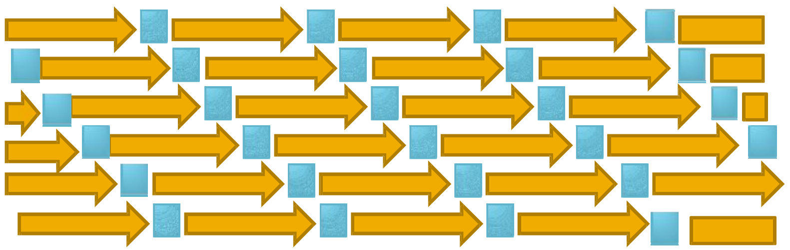


FORMATION OF FIBRILS

Tropocollagen $\longrightarrow$ Collagen fibril

The way of aggregation of fibrillary collagen

- Regular arrangement along the row and in the adjacent row
- The adjacent row is displaced by $\frac{1}{4}$ of the length



ASSEMBLING OF COLLAGEN FIBRILS

Polymerisation

Tropocollagen

↓

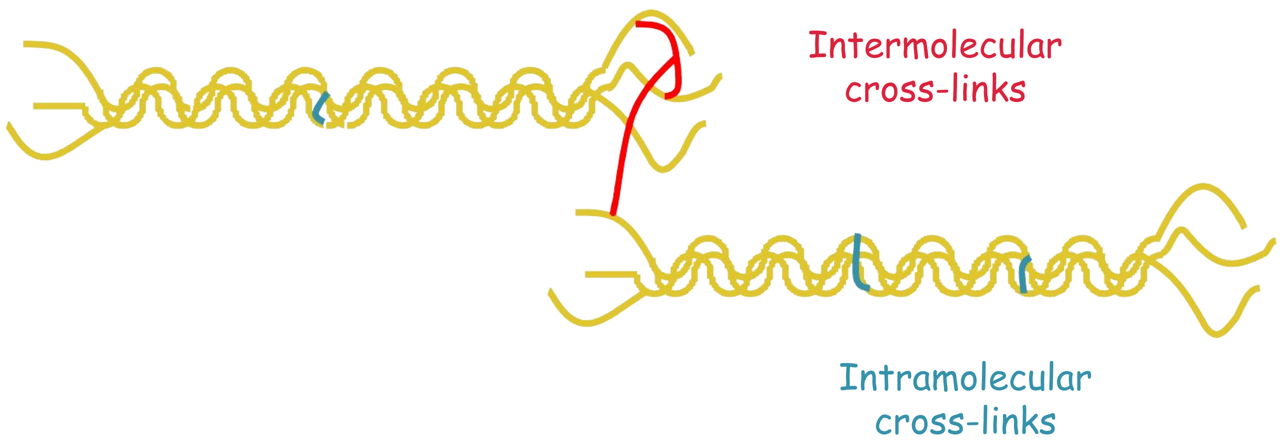
Collagen fibrils

↓

Collagen fibers

FORMATION OF CROSS-LINKS

Collagen fibers are stabilized by formation of the covalent cross-links, which can be formed either within the tropocollagen molecule between the three chains – intramolecular cross-links and between the tropocollagen molecules – intermolecular cross-links.



FORMATION OF CROSS-LINKS

Function of cross-links

stabilization and strengthening of collagen fibril

$\downarrow$ Cross-linking

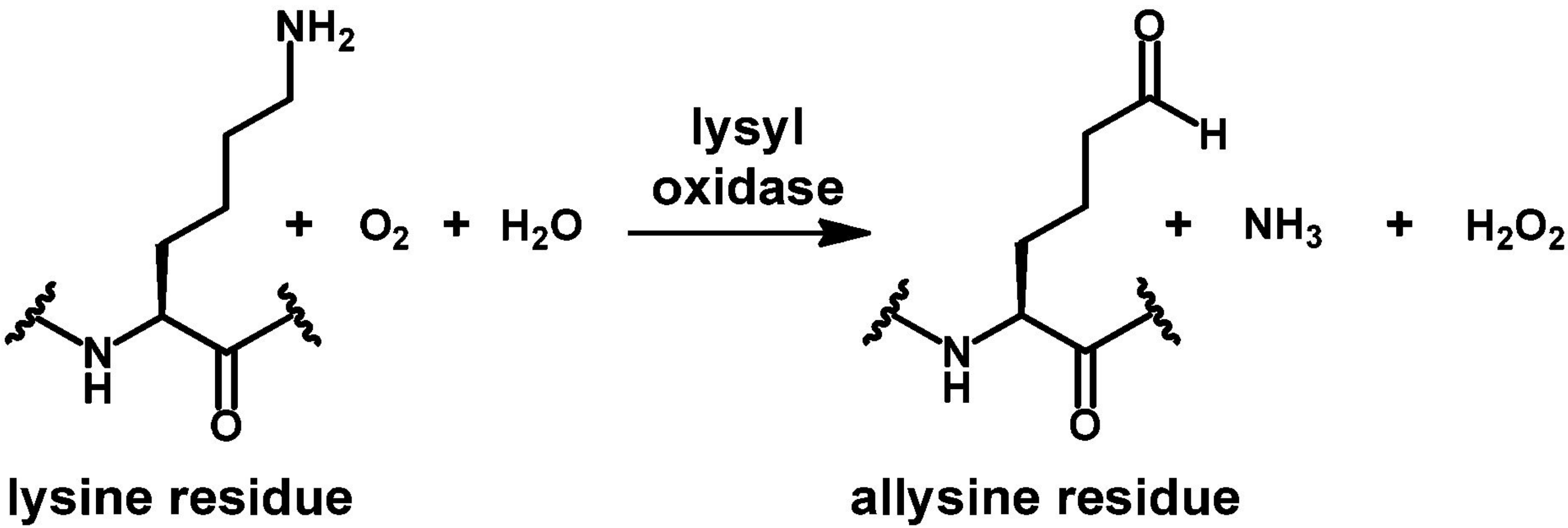
high breaking strength
lower extensibility

ALDOL CROSS-LINK

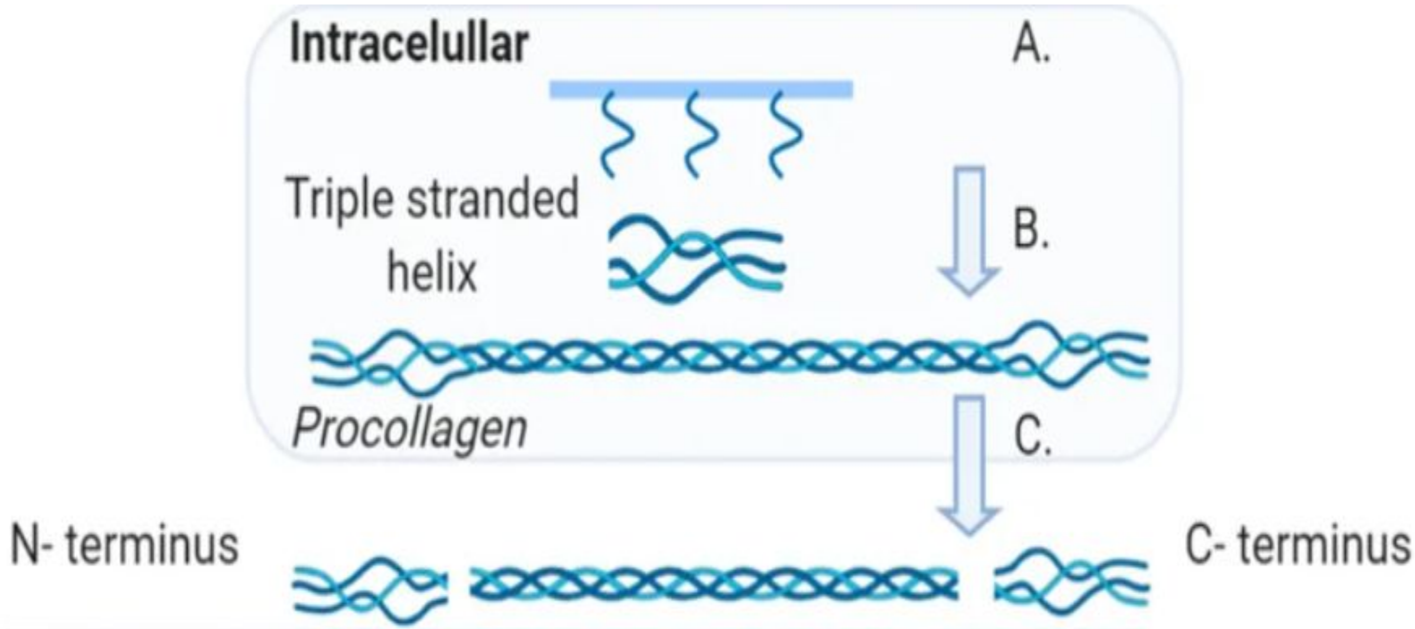
- Mechanism of formation

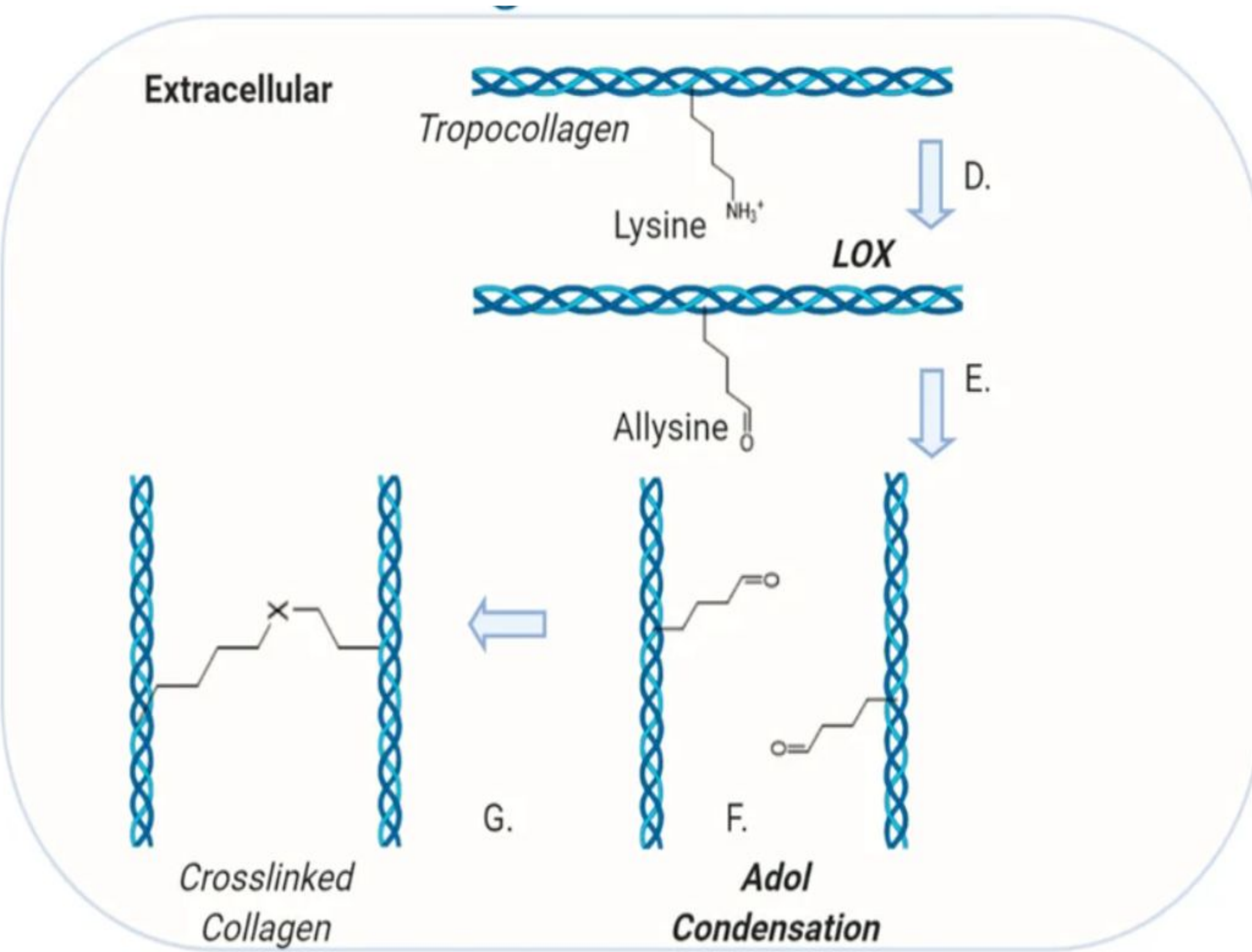
Oxidative deamination of lysine, aldehyde formation

- by the enzyme lysyloxidase
- aminooxidase, containing Cu^{2+}
-

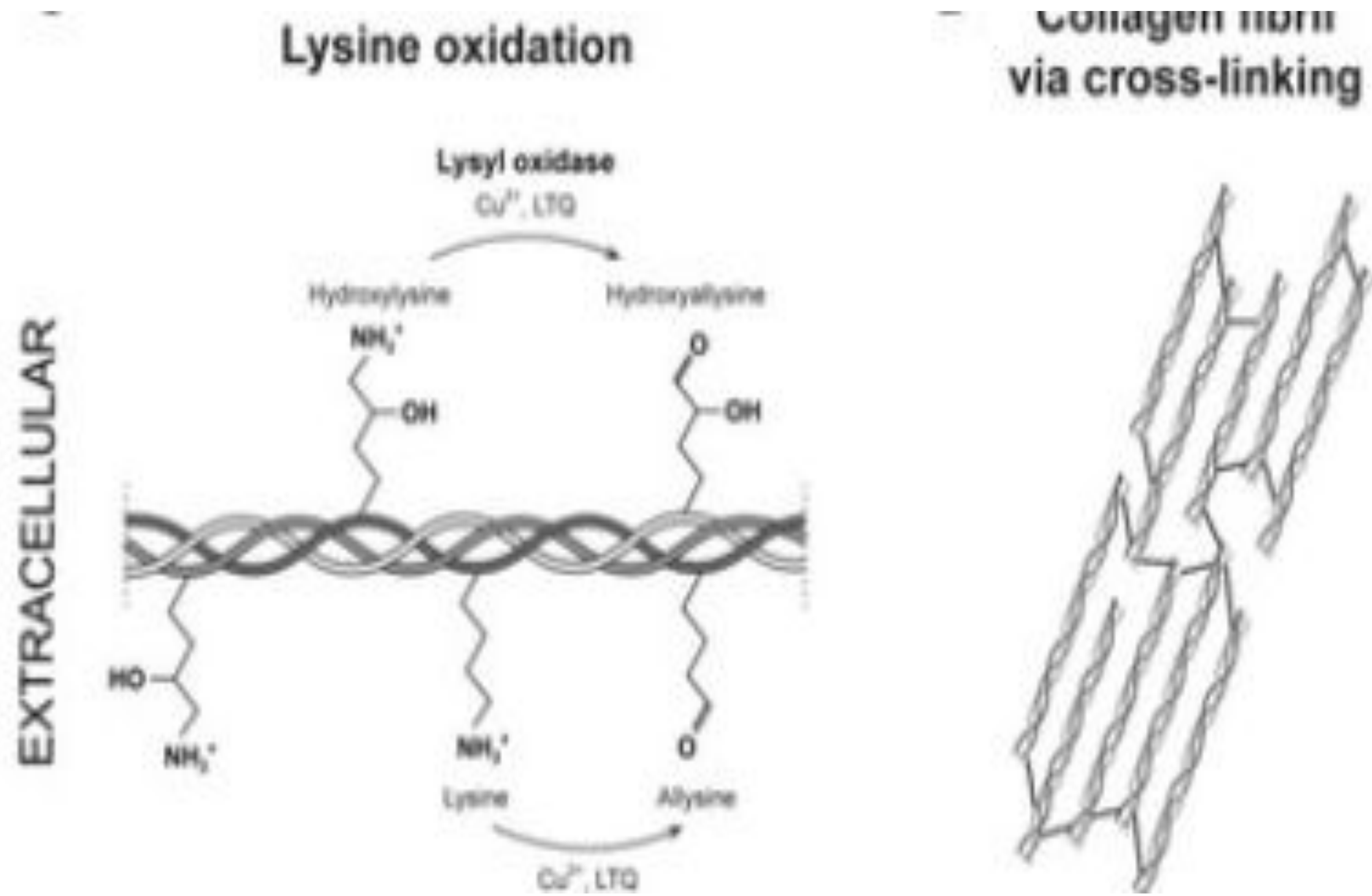


NH_2
+ O_2 + H_2O
lysyl
oxidase
O
H
+ NH_3 + H_2O_2
N
H
O
N
H
O
lysine residue
allysine residue


Intracelullar
A.
Triple stranded
helix
B.
Procollagen
C.
N- terminus
C- terminus



Extracellular
Tropocollagen
D.
Lysine
NH_3^+
LOX
E.
Allysine
O
O
O
Crosslinked
Collagen
G.
F.
Adol
Condensation



Lysine oxidation
via cross-linking
Lysyl oxidase
Cu^{2+}, LTQ
Hydroxylysine
Hydroxyallysine
NH_3^+
OH
O
OH
EXTRACELLULAR
HO
NH_3^+
NH_3^+
O
Lysine
Allysine
Cu^{2+}, LTQ

OVERVIEW OF COLLAGENS

Some fibrilar collagens

Type	Molecular	Occurrence structure
I	$[\alpha_1(I)]_2\,[\alpha_2(I)]$	widely present, skin, vessels, tendons, gingiva, bone, cement, dentin, periodontal ligaments
II	$[\alpha_1(II)]_3$	cartilage, vitreous body
III	$[\alpha_1(III)]_3$	skin, vessels, lungs, gingiva, cement, dentin, periodontal ligaments
V	$[\alpha_1(V)]_3$, $[\alpha_1(V)_2\,\alpha_2(V)]$	skin, smooth muscle, bone, cement, dentin

TYPES OF COLLAGEN

Some collagens associated with fibrils

Type	Molecular structure	Occurrence
VI	$[\alpha_1(VI)\ \alpha_2(VI)\ \alpha_3(VI)]$	laterally associated with collagen type II, widely present, bone, gingiva, cement, periodontal ligaments
IX	$[\alpha_1(IX)\ \alpha_2(IX)\ \alpha_3(IX)]$	laterally associated with collagen type II, cartilage, vitreous body, periodontal ligaments
XII	$[\alpha_1(XII)]_3$	associated with collagen type I in soft tissues, periodontal ligaments

OVEVIEW OF COLLAGENS

Some net forming collagens

Type	Molecular	Occurrence	structure
IV	$[\alpha_1(IV)]_2\,[\alpha_2(IV)]$	basal membranes, formation of two-dimensional net gingiva, periodontal ligaments	

DISORDERS OF COLLAGEN SYNTHESIS

Increased collagen synthesis

- fibroses

Decreased collagen synthesis

- genetic disorder
- acquired disorders

DISORDERS OF COLLAGEN SYNTHESIS

Increased collagen synthesis - FIBROSIS

- Hepatic cirrhosis
- Pulmonary fibrosis
- Atherosclerosis

Tissue damage stimulates collagen synthesis by fibroblasts

- e.g. damaged hepatocytes are replaced by fibrous connective tissue hepatic cirrhosis

DISORDERS OF COLLAGEN SYNTHESIS

Decreased collagen synthesis

- Genetically conditioned
  - Ehlers-Danlos syndrome
  - osteogenesis imperfecta

- Acquired disturbances
  - Copper deficiency
  - vitamin C deficiency

ELASTIN IN ECM

Elastin is the main protein of elastic fibers, providing elasticity and recoil to the tissues.

Found in tissues requiring flexibility.

Contains desmosine cross-links.

Less abundant in dental hard tissues but important in soft tissues.

ELASTI
N

Occurrence

- Arteries, particularly in aorta
- Skin
- Elastic tendons
- Lungs

ELASTIN

Properties

Extensibility and contractility

- Resembles the rubber
- After extension elastin is able to return to original size and original form
- Tensile strength is lower than in collagen
- Hydrophobic, practically insoluble in aqueous solutions

Amino acids:

- Small and nonpolar (e.g., glycine, alanine, and valine).
- Also rich in proline and lysine but contains few hydroxyproline and hydroxylysine.
- *Lysine takes part in forming of cross - links.*

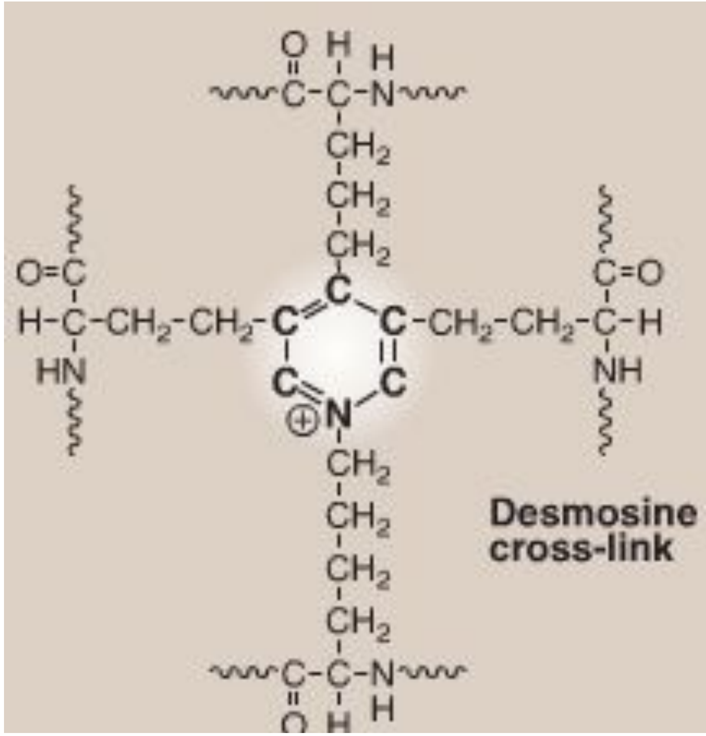
CROSS-LINKS IN ELASTIN

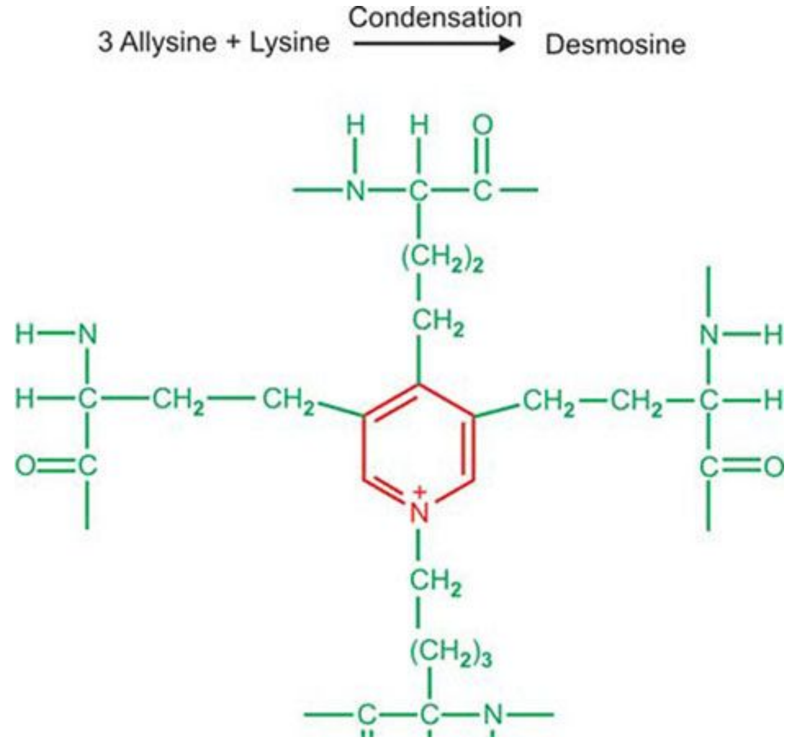
- Key step is an oxidative deamination of some lysine residues by copper-containing lysyloxidase (the same enzyme as in formation of cross-links in collagen).
- Cross-links may be formed within one polypeptide chain or between 2 – 4 chains

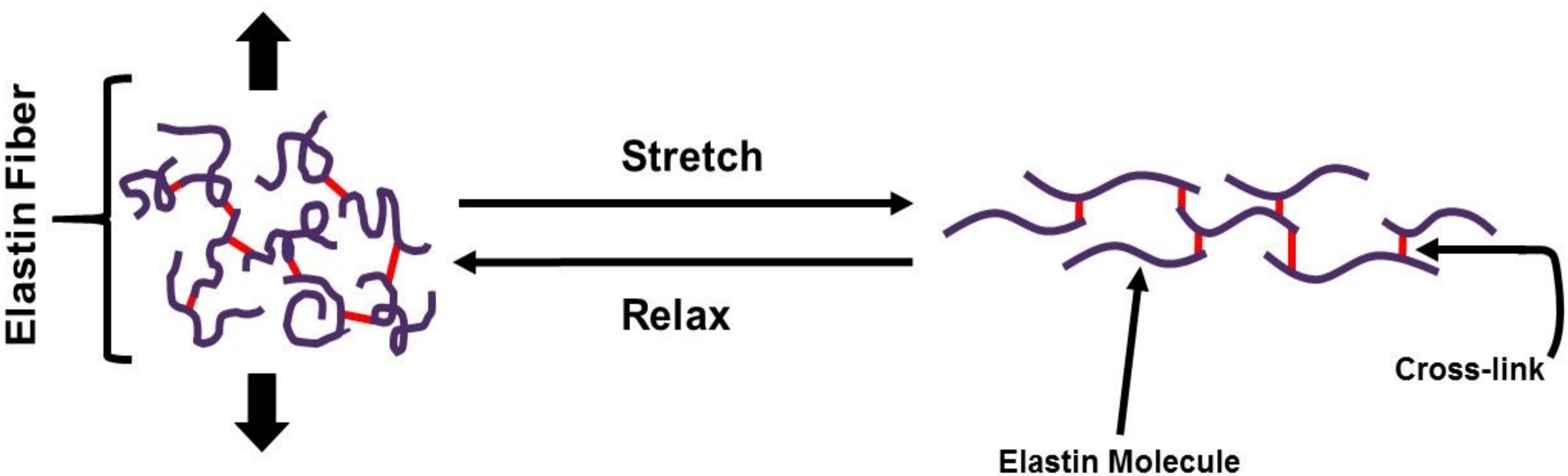
DESMOSINE

- Cross-link completely specific for elastin
- Arises from 4 side chains of LYSINE (3 oxidized and 1 nonoxidized)
- Determines the high elasticity of elastin

Linking of polypeptide chains of elastin by cross-links constitutes a three-dimensional netting explaining the “rubber-like” properties of elastin.


Desmosine
cross-link


3 Allysine + Lysine
Condensation
Desmosine


Elastin Fiber
Stretch
Relax
Cross-link
Elastin Molecule

ELASTIN SYNTHESIS

Intracellular processes

SYNTHESIS OF POLYPEPTIDE CHAIN

HYDROXYLATION OF PROLINE

RESIDUES SECRETION OF

TROPOELASTIN

Extracellular processes

Tropoelastin

(globular structure, Mr = 70 000)

Formation of cross-links

Three-dimensional netting

THANK YOU